

pandas.DataFrame.value_counts

We use
"DataFrame.value_counts"
to count the frequency of
each unique row in the
DataFrame.

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor



Manager	Product	
Debra	CPU	2
Fred	Monitor	2
John	Software	1

Name: count, dtype: int64

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1st Edition

Why This E-book?

"The aim of this ebook is to give you the 'aha' moment right away at the start of learning a new concept."

- Practical step By Step Guide With Simple Examples
- Visual Illustrations and Interactive
- Simple Datasets
- Comprehensive Coverage (pandas Documentation used as reference)

Tips for Effective Use of This Ebook

Use Examples First: If you don't understand the text, go straight to the examples—they're self-explanatory.

After reviewing them, return to the text to grasp the practical concepts.

Practice with Datasets: Copy the datasets provided and practice using them to reinforce your understanding.

Click on the blue links to go directly to the section you want to learn about.

Introduction

```
#import Libraries
import numpy as np
import pandas as pd
```

We use

`"DataFrame.value_counts"`
to count the frequency of
each unique row in the
`DataFrame`.

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor



Manager	Product	
Debra	CPU	2
Fred	Monitor	2
John	Software	1

Name: count, dtype: int64

Code Snippet:

```
data = {  
    'Manager': ['Debra', 'Debra', 'Fred', 'John', 'Fred'],  
    'Product': ['CPU', 'CPU', 'Monitor', 'Software', 'Monitor']  
}  
df = pd.DataFrame(data)  
df
```

df:

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor

The subset parameter

The subset parameter accepts a label or a list of labels, and it is optional.
It Specifies Columns to use when counting unique combinations.

How many times does each product appear in the DataFrame, regardless of the manager?

When
"subset"
accepts single
column label
(like 'Product')

df:

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor



Output:

Product	
CPU	2
Monitor	2
Software	1

Name: count, dtype: int64

Code Snippet:

```
df.value_counts(subset=['Product'])
```

Code Snippet:

```
data = {  
    'Manager': ['Debra', 'Debra', 'Fred', 'John', 'Fred'],  
    'Product': ['CPU', 'CPU', 'Monitor', 'Software', 'Monitor']  
}  
  
df = pd.DataFrame(data)  
df
```



df:

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor

How many times does each unique combination of product and price appear in the DataFrame, disregarding the manager?

When
"subset"
accepts
a list of
column labels.
(like ['Product',
'Price']).

df:

	Manager	Product	Price
0	Debra	Monitor	300
1	Debra	CPU	600
2	Fred	Monitor	300
3	John	Monitor	480
4	Fred	Monitor	300



Output:

```
Product  Price    count
Monitor   300         3
CPU        600         1
Monitor   480         1
Name: count, dtype: int64
```

Code Snippet:

```
df.value_counts(['Product', 'Price'])
```

Code Snippet:

```
data = {
    'Manager': ['Debra', 'Debra', 'Fred', 'John', 'Fred'],
    'Product': ['Monitor', 'CPU', 'Monitor', 'Monitor', 'Monitor'],
    'Price': [300, 600, 300, 480, 300]
}

df = pd.DataFrame(data)
df
```



df:

	Manager	Product	Price
0	Debra	Monitor	300
1	Debra	CPU	600
2	Fred	Monitor	300
3	John	Monitor	480
4	Fred	Monitor	300

How many times does each combination of 'Manager' and 'Product' occur in the DataFrame?

without subset:

(Default behavior):

It will count

Unique

Combinations of

Both Columns

df

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor



Ouput:

```
Product Price count
Monitor 300      3
CPU      600     1
Monitor 480     1
Name: count, dtype: int64
```

Code snippet:

```
df.value_counts()
```

df:

Manager Product

0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor

Code Snippet:

```
data = {
    'Manager': ['Debra', 'Debra', 'Fred', 'John', 'Fred'],
    'Product': ['CPU', 'CPU', 'Monitor', 'Software', 'Monitor']
}
df = pd.DataFrame(data)
df
```

The normalize parameter

The "normalize" parameter returns proportions rather than frequencies.

The "normalize" parameter accepts Boolean, and defaults to False.

How many times does each unique combination of 'Manager' and 'Product' occur in the DataFrame?

df:

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor

(normalize=False):
(without subset):
(Default behavior)

→

Output:

Manager	Product	
Debra	CPU	2
Fred	Monitor	2
John	Software	1

Name: count, dtype: int64

Code Snippet:

```
df.value_counts(normalize=False)
```

df:

	Manager	Product
0	Debra	CPU
1	Debra	CPU
2	Fred	Monitor
3	John	Software
4	Fred	Monitor

When
normalize = True

→

Output:

Manager	Product	
Debra	CPU	0.4
Fred	Monitor	0.4
John	Software	0.2

Name: proportion, dtype: float64

Code Snippet:

```
df.value_counts(normalize = True)
```

The **sort** parameter

The "sort" parameter accepts Boolean, and defaults to True.

It Sorts by frequencies when True

Sort by DataFrame column values when False.

When sort=True (default):

It sorts by

Sort by frequencies

(most frequent first).

df:

	fruits
0	apple
1	banana
2	apple
3	orange
4	banana
5	banana



Output:

```
fruits
banana    3
apple     2
orange    1
Name: count, dtype: int64
```

Code Snippet:

```
df.value_counts(sort = True)
or
df.value_counts()
```

When sort=False:

Sort by

DataFrame column values.

df:

	fruits
0	apple
1	banana
2	apple
3	orange
4	banana
5	banana



Output:

```
fruits
apple    2
banana   3
orange   1
Name: count, dtype: int64
```

Code Snippet:

```
df.value_counts(sort=False)
```

Code Snippet:

```
df = pd.DataFrame({  
    'fruits': ['apple', 'banana', 'apple', 'orange', 'banana', 'banana']  
})  
df
```

→

df:	
	fruits
0	apple
1	banana
2	apple
3	orange
4	banana
5	banana

The ascending parameter

The "ascending" parameter accepts Boolean, and defaults to False.

It controls the order in which these counts are sorted.

df:

	fruits
0	apple
1	banana

Output:

fruits
banana 3

Code Snippet:

When ascending=False (default):

It sorts from
largest to smallest .

	fruits
2	apple
3	orange
4	banana
5	banana



apple 2
orange 1
Name: count, dtype: int64

df.value_counts(ascending=False)
or
df.value_counts()

When ascending = True:
It sorts from
smallest to largest.

df:

	fruits
0	apple
1	banana
2	apple
3	orange
4	banana
5	banana



Output:

fruits
orange 1
apple 2
banana 3
Name: count, dtype: int64

Code Snippet:

df.value_counts(ascending=True)

The `dropna` parameter

The "dropna" parameter accepts Boolean, and defaults to True.

It doesn't count rows that have missing values."

df:

Output:

When dropna=True (Default Behavior):
Rows that contain NA values are
not counted.

	A
0	1.0
1	2.0
2	NaN
3	2.0
4	3.0
5	NaN
6	3.0



A
2.0 2
3.0 2
1.0 1
Name: count, dtype: int64

Code Snippet:

```
df.value_counts(dropna=True)  
or  
df.value_counts()
```

When dropna=False,
rows that contain NA values are
included in the counts.

df:

	A
0	1.0
1	2.0
2	NaN
3	2.0
4	3.0
5	NaN
6	3.0



Output:

2.0 2
3.0 2
NaN 2
1.0 1
Name: count, dtype: int64

Code Snippet:

```
df.value_counts(dropna=False)
```

Code Snippet:

```
data = {'A': [1, 2, np.nan, 2, 3, np.nan, 3]}
df = pd.DataFrame(data)
df
```

→

A	
1	2.0
2	NaN
3	2.0
4	3.0
5	NaN
6	3.0

```
%%html
```

```
<h2 style="text-align: center; color: blue" id="sources">Sources & References</a><br>
```

```
<div style="background-color: #f0f0f0; padding: 10px; border-radius: 8px; box-shadow: 0 0 10px rgba(0,0,0,0.1); font-family: Ar
  <p><a href="https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.value_counts.html" target="_blank" style="color:
    pandas.DataFrame.value_counts Documentation</a> (<span style="font-family: monospace;">https://pandas.pydata.org/docs/refer
  </div>
```

Sources & References

[pandas.DataFrame.value_counts Documentation](https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.value_counts.html)

(https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.value_counts.html)

Other E-books on pandas

Download (https://github.com/kicherethedatascientist/My_e-Books)

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Kichere Magubu is a data enthusiast and content creator.

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Thank You!

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```
#print("The cell to convert jupyter notebook to html")  
!jupyter nbconvert --to hide_code_html "value_counts.ipynb"
```